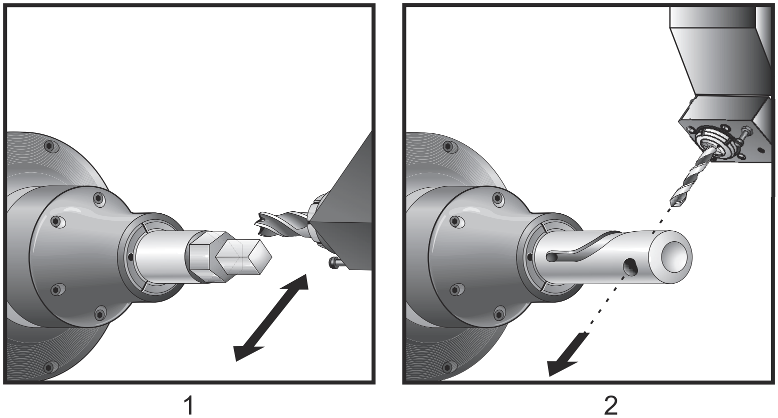
6.6 Live Tooling

This option is not field installable.

F6.6: Axial and Radial Live Tooling: [1] Axial Tool, [2] Radial Tool.



6.6.1 Live Tooling Introduction

The live tooling option allows the user to drive axial or radial tools to perform such operations as milling, drilling, or slotting. Milling shapes is possible using the C-Axis and/or the Y Axis.

Live Tooling Programming Notes

The live tool drive automatically turns itself off when a tool change is commanded.

For the best milling accuracy, use the spindle clamp M Codes (`M14`- Main Spindle / `M114` - Secondary Spindle) before machining. The spindle automatically unclamps when a new main spindle speed is commanded or**[RESET]** is pressed.

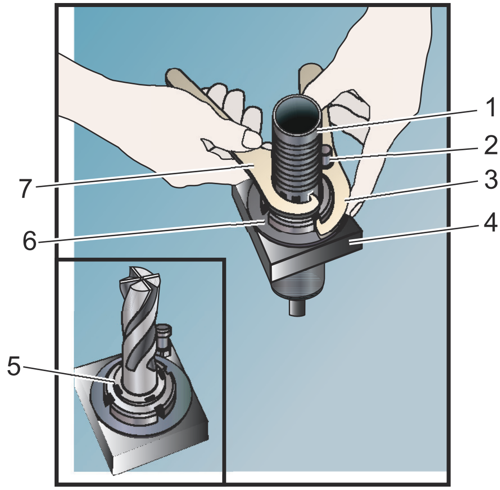
Maximum live tooling drive speed is 6000 RPM.

Haas live tooling is designed for medium duty milling, e.g.: 3/4" diameter end mill in mild steel max.

6.6.2 Live Tooling Cutting Tool Installation

CAUTION: *I Never tighten the live tool collets on the turret. Tightening a live tool collet that is on the turret will cause damage to the machine.*

F6.7: ER-32-AN Tube Wrench and Spanner: [1] ER-32-AN Tube wrench, [2] Pin, [3] Spanner 1, [4] Tool holder, [5] ER-32-AN nut insert, [6] Collet housing nut, [7] Spanner 2.



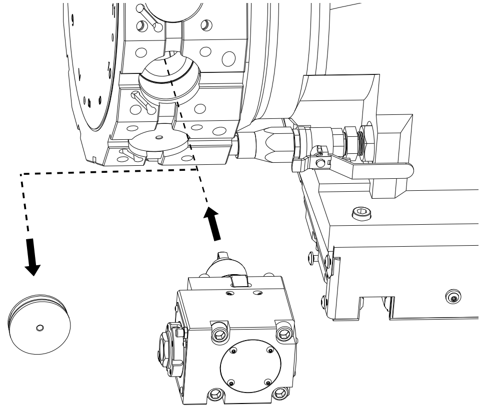
1. Insert the tool bit into the ER-AN nut insert. Thread the nut insert into the collet housing nut.
2. Place the ER-32-AN tube wrench over the tool bit and engage the teeth of the ER-AN nut insert. Tighten the ER-AN nut insert by hand using the tube wrench.
3. Place Spanner 1 [3] over the pin and lock it against the collet housing nut. It may be necessary to turn the collet housing nut to engage the spanner.
4. Engage the teeth of the tube wrench with Spanner 2 [7] and tighten.

6.6.3 Live Tool Mounting in Turret

To mount and install live tools:

1. Mount a radial or axial live tool holder and snug the mounting bolts.
2. Torque the mounting bolts in a criss-cross pattern to 60 ft-lbs (82 N-m). Make sure the bottom face of the toolholder is clamped flush with the face of the turret.

F6.8: Live Tool Installation



6.6.4 Live Tooling M-codes

The following M-Codes are used in Live Tooling. Also, refer to the M-codes section starting on page **365**.

M19 Orient Spindle (Optional)

`M19` adjusts the spindle to a fixed position. The spindle only orients to the zero position without the optional `M19` orient spindle feature.

The orient spindle function allows `P` and `R` address codes. For example, `M19 P270.` orients the spindle to 270 degrees. The `R` value allows the programmer to specify up to two decimal places; for example, `M19 R123.45`. View the angle in the **Current Commands Tool Load** screen.

`M119` positions the secondary spindle (DS lathes) the same way.

Spindle orientation is dependent on the mass, diameter, and length of the workpiece and/or the workholding (chuck). Contact the Haas Applications Department if any unusually heavy, large diameter, or long configuration is used.

M219 Live Tool Orient (Optional)

P - Number of degrees (0 - 360)
R - Number of degrees with two decimal places (0.00 - 360.00).

`M219` adjusts the live tool to a fixed position. An `M219` orients the spindle to the zero position. The orient spindle function allows P and R address codes. For example:

```
M219 P270. (orients the live tool to 270 degrees) ;
```

The `R`-value allows the programmer to specify up to two decimal places; for example: